



Assignment #1

DS-3003

DATA WAREHOUSING & BUSINESS INTELLIGENCE

DEADLINE: 08 October 2024, 5pm PST

Total Marks: 100

Weightage: 5%

Important Instructions

- Try to complete the assignment before the deadline.
- Try to attempt all questions! Data warehouse is a highly useful and fast-growing field. This assignment will help you understand the basics well.
- Plagiarism is strictly prohibited! Try to solve the assignment on your own. Marks will be given based on your thought process so make sure you have a solid reason for all your attempts.
- The assignment is of 100 marks in total. Individual marks for each question and part are mentioned below alongside the questions.
- The assignment should be written in a document (ms-word or google-docs) and submitted on Google Classroom (GCR).
- You can use Whimsical, Miro, or Draw.io to create diagrams

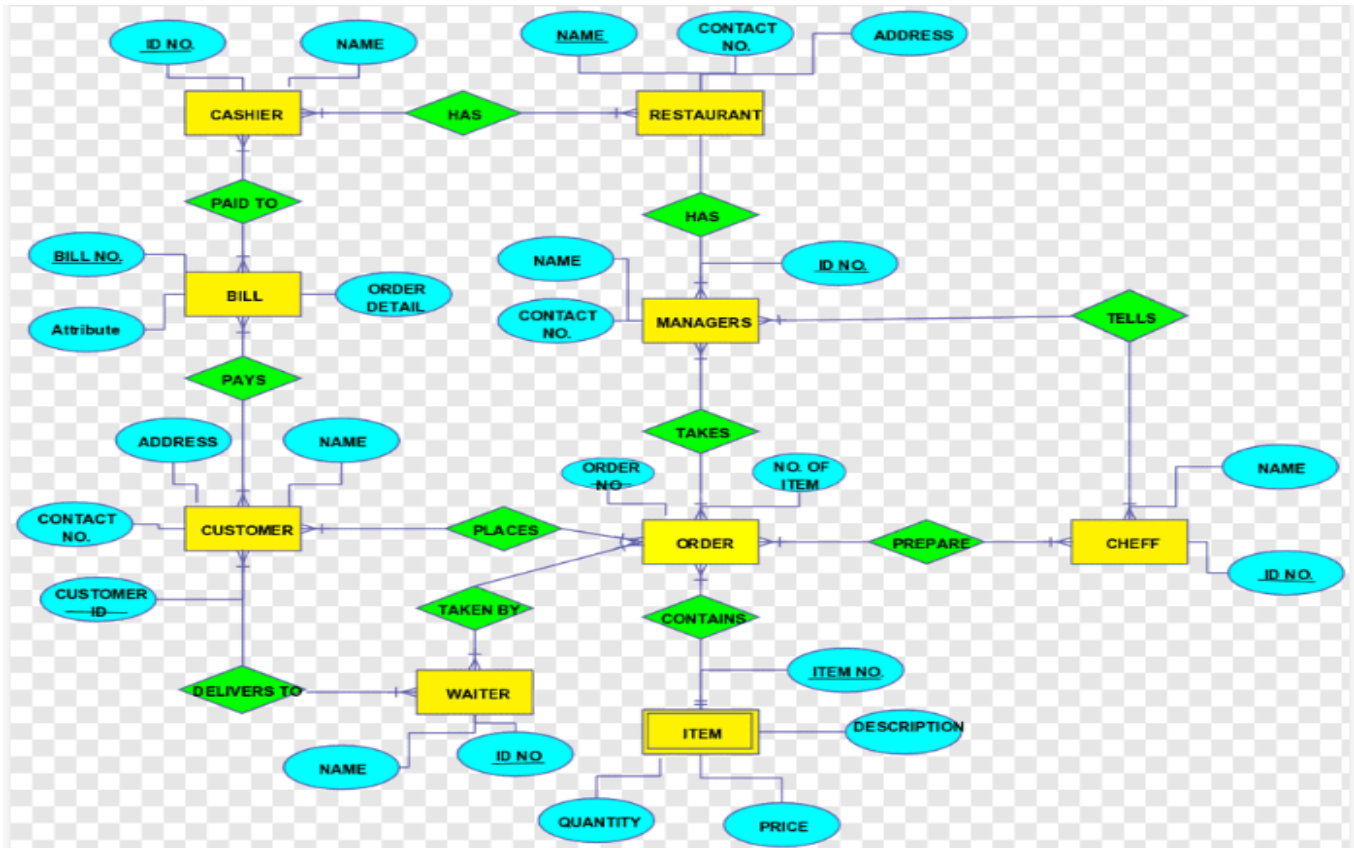
QUESTION: 01

James Anderson is the founder of "GreenGadget," a company that manufactures and sells eco-friendly tech gadgets across various countries in Europe. The company operates 20 stores and has an online presence as well. Recently, GreenGadget has seen a significant dip in customer retention and sales. James wants to leverage a data warehouse to gain deeper insights into the sales trends and customer behavior. (35 marks)

- Create a Multi-Dimensional Entity Relationship (MER) model for GreenGadget. **(10 marks)**
- Describe the storage structure that would be most suitable for GreenGadget's data warehouse. Should James use a relational database (RDB) or a multidimensional database (MDB), or a combination of both? Justify your answer based on the needs of the business. **(10 marks)**
- What are some important dimensions and facts James should focus on when analyzing customer satisfaction and delivery efficiency? Discuss how these insights could help improve GreenGadget's performance. **(8 marks)**
- Briefly describe the ETL (Extraction, Transformation, Loading) process for GreenGadget's data warehouse, outlining how it would be applied to integrate data from their stores and online sales channels. **(7 marks)**

QUESTION: 02

Given below is the ERD for a hotel management system. Study it carefully and then solves the questions below for a multidimensional database. (25 marks)



- We know that a Data Warehouse is subject-oriented. Give 3 examples of subjects which may be suitable for the above hotel management system. (2.5 marks)
- Discover and create all fact tables with attributes you think are appropriate. (10 marks)
- Discover and create all dimension tables with attributes you think are appropriate. (10 marks)
- Identify 2 possible measures that could be useful for analyzing the performance of the hotel. Justify your choices. (2.5 marks)

QUESTION: 03

Sketch and explain the architecture of a Data Warehouse for a 2-tier, 3-tier, and a 5-tier architecture: (15 marks)

- Sketch the 2-tier architecture of a Data Warehouse. For each tier, explain its role and function within the architecture. (05 marks)
- Sketch the 3-tier architecture of a Data Warehouse. For each tier, explain its role and function within the architecture. (05 marks)
- Sketch the 5-tier architecture of a Data Warehouse. For each tier, describe what it represents and how it contributes to the overall data flow and decision support system. (05 marks)

QUESTION: 04

FreshHarvest is a global agricultural company specializing in the production and distribution of organic food products. With farms located in North America, South America, Europe, and Asia, each region faces distinct agricultural conditions and market demands. FreshHarvest aims to enhance its operational efficiency and profitability by leveraging its data effectively. The management is looking to optimize their data warehousing and business intelligence capabilities to better understand market trends, manage supply chains, and improve customer relationships. (25 marks)

Keeping in mind the above scenario, solve the following questions on data warehouse for FreshHarvest:

1. Data Warehouse Design: (10)

- How would you set up the data storage for FreshHarvest Store considering it operates in different regions? Explain your approach.
- What are some specific details that would change in the data storage depending on the region?

2. Data Warehouse Operations: (10)

- What's the best way to collect data in this scenario? Explain why you choose this method.
- When you're converting data from different branches and combining it into the data storage, what are five problems that might come up, and how would you solve them?

3. Bulk Loading Time: (2.5)

- When would be the most convenient time to load a large amount of data into the system?

4. Business Analysis: (2.5)

- For business analysis and getting useful information, name five important things FreshHarvest Store should be able to find out from its data. It can be in the form of questions they should be able to answer.

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